Project Title: SafetyXperience
Type: VR Emergency Training Simulator
Engine: Unity
Platform: Meta Quest 3
Deadline: 10 March

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1. Project Overview

SafetyXperience is a **VR-based emergency training simulator** designed to teach users how to respond correctly in real-life emergency situations.

The game will contain **multiple interactive emergency scenarios**, each teaching practical safety steps in a gamified way.

The final product must be:

- Fully functional
- Polished
- VR optimized
- Step-based
- Educational
- Clean and organized

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2. Core Game Structure

Main Menu

The game must include:

- Game Title
- Scenario Selection
- Settings
- Exit

Scenario Selection Screen

Player can choose between:

1. Earthquake Scenario
2. Fire Emergency Scenario
3. Flood Scenario

Each scenario should load separately.

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3. Scenario 1 – Earthquake Rescue

Flow:

1. Earthquake starts.
2. Player must find cover (safe zone).
3. Earthquake stops.
4. Player hears trapped NPC.
5. Player picks tool.
6. Player clears debris.
7. Player rescues NPC.
8. Feedback screen appears.

Required Systems:

- Camera shake
- Safe zone detection
- Tool pickup (VR interaction)
- Breakable obstacle
- NPC interaction
- Timer
- Score system
- Feedback UI

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4. Scenario 2 – Fire Emergency

Flow:

1. Fire alarm triggers.
2. Player navigates to find fire extinguisher
3. Player finds the source of fire.
4. Player must:
    - Pull pin.

    - Aim at fire base.
    - Extinguish fire.
5. Exit safely.
6. Feedback screen.

Required Systems:

- Fire particle system
- Extinguisher interaction
- Fire suppression mechanic
- Objective system
- Timer & scoring
- Feedback screen

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5. Scenario 2 – Flood Emergency (Using Cesium)

Technology Requirement

This scenario must use **Cesium for Unity**

Cesium will be used to create a real-world geospatial flood simulation environment.

Scenario Overview

This scenario simulates a **realistic flood situation in a real-world location** using Cesium's 3D terrain and map data.

The objective is to train the player to:

- Move to higher ground
- Avoid electrical hazards
- Assist a stranded NPC
- Reach a designated safe evacuation zone

Environment Setup (Cesium-Based)

Must:

1. Integrate Cesium into Unity.

2. Load a real-world terrain location (can be any location of Islamabad).
3. Use Cesium World Terrain.
4. Use Cesium 3D Tiles for buildings (if possible).
5. Keep it optimized for VR.

- Environment does NOT need to be highly detailed (functionality over realism).

Scenario Flow

Phase 1 – Warning

- Scenario begins in a normal outdoor environment.
- Audio alert: Flood warning.
- Objective appears:
  "Move to higher ground immediately."

Water level begins rising slowly.

Phase 2 – Rising Water

- Water level increases gradually.
- Certain areas become hazardous.
- Player must:
  - Avoid flooded electric zones.
  - Avoid staying in low areas too long.
- Timer tracks response speed.

If player remains in danger zone too long → penalty.

Phase 3 – Rescue

- NPC is stranded on slightly elevated terrain.
- Player must:
  - Reach NPC safely.
  - Interact to assist.
  - Escort NPC to safe evacuation point.

Simple interaction system is enough (grab or trigger interaction).

Phase 4 – Completion

When both player and NPC reach safe zone:

Display Feedback Screen with:

- Time taken
- Hazard exposure time
- Mistakes made
- Final score

Buttons:

- Restart Scenario
- Return to Main Menu

Must be VR interactive.

Required Systems

System	Requirement
Cesium Integration	Properly working terrain
Water System	Rising water effect
Hazard Zones	Electric areas or dangerous zones
Safe Zone Detection	High ground detection
NPC Escort System	Basic follow system
Timer	Tracks completion time
Score System	Based on performance
Feedback Screen	VR compatible

Technical Notes for Developer

- Optimize Cesium tile loading for VR.
- Limit view distance to maintain performance.
- Avoid heavy 3D building tiles if causing lag.
- Use simple water shader for rising effect.

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6. Core Gameplay Systems (Must Be Used Across All Scenarios)

Objective System

- On-screen objective text
- Step progression

- Cannot skip steps

Scoring System

Score based on:

- Time taken
- Mistakes
- Correct actions

Feedback Screen (For Every Scenario)

Display:

- Time taken
- Score %
- Mistakes made
- Performance message

Buttons:

- Restart
- Next Scenario
- Return to Menu

Must be VR interactive.

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7. UI Requirements

- VR-compatible UI
- Clean layout
- World-space canvas
- No complex animations required

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8. Sound Requirements

- Background ambient sounds
- Emergency sounds (earthquake, fire, water)
- NPC voice

- UI click sounds

All assets must be:

- Royalty free
- Optimized

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9. Technical Requirements

- Unity (latest stable LTS)
- Organized scripts
- Clean project hierarchy
- Optimized models
- No copyrighted content

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10. Deliverables Required

Freelancer must provide:

1. Complete Unity Project Folder
2. All C# scripts organized
3. Working VR Build
4. Demo video of each scenario
5. Clean folder structure
6. Basic documentation explaining:
    - Scenario logic
    - Scoring system
    - Interaction system

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11. Optimization Requirements

- Smooth performance in VR
- No lag spikes
- No heavy textures
- No high-poly models

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12. Project Milestones

Stage	Work	Expected Time
Stage 1	Main menu + framework	2 days
Stage 2	Earthquake scenario	4–5 days
Stage 3	Fire scenario	3–4 days
Stage 4	Flood scenario	3–4 days
Stage 5	UI polish + bug fixing	2 days

Total estimated time: 12–15 days

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14. Important Conditions

- Daily progress updates required
- Short demo video daily
- No disappearing without updates
- Clear communication
- Code must be understandable

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15. Quality Expectation

This is an academic VR simulator. It does NOT need AAA graphics.

Priority:

1. Functionality
2. Clear learning steps
3. Smooth VR interaction
4. Clean UI